



UNIVERSITY INSTITUTE OF COMPUTING

PROJECT REPORT ON ONLINE LEARNING BEHAVIOUR ANALYSIS

Program Name: BCA(Data Science)

Subject Name/Code: Data Mining Lab

24CAP-292

Submitted by:	Submitted to:
Name: Ashmita	Name: Ms. Ankita
UID: 24BCD10012	Designation: Assistant Professor
Section: 24BCD1-(A)	



ONLINE LEARNING BEHAVIOUR ANALYSIS

1. Introduction

The rapid expansion of digital technologies and internet accessibility has significantly transformed the education sector, leading to the widespread adoption of online learning platforms. These platforms generate a massive amount of data through student interactions such as video watch time, quiz attempts, login frequency, course completion rates, and engagement levels.

Data mining plays a crucial role in extracting meaningful information from such large datasets. By applying various analytical techniques, hidden patterns, relationships, and trends can be discovered, which are not easily visible through traditional analysis. In this project, WEKA (Waikato Environment for Knowledge Analysis), a powerful and user-friendly data mining tool, is used to analyze an online learning dataset.

The project focuses on applying classification, clustering, association rule mining, and visualization techniques to understand student behaviour. The insights generated from this analysis can help educators and institutions improve course design, enhance student engagement, identify at-risk learners, and ultimately improve learning outcomes.

2. Objectives of the Study

- To analyze student behaviour in an online learning environment by examining interaction data such as watch time, quiz performance, and login frequency in detail.
- To identify key factors that significantly influence student performance, engagement levels, and course completion rates.
- To classify students into performance categories (High, Medium, Low) using classification techniques for better understanding and prediction.
- To group students into clusters based on similarity in behaviour and learning patterns using clustering techniques.
- To discover meaningful relationships and hidden patterns among different attributes using association rule mining.

3. Dataset Description

The dataset consists of 50 student records with attributes representing their activity and performance in online courses. These attributes include:

The dataset consists of 50 student records with the following attributes:

- Student_ID, Course, Watch_Time, Quiz_Score, Completion_%, Login_Frequency, Drop_Off, Performance Category, Engagement Level, Risk Category

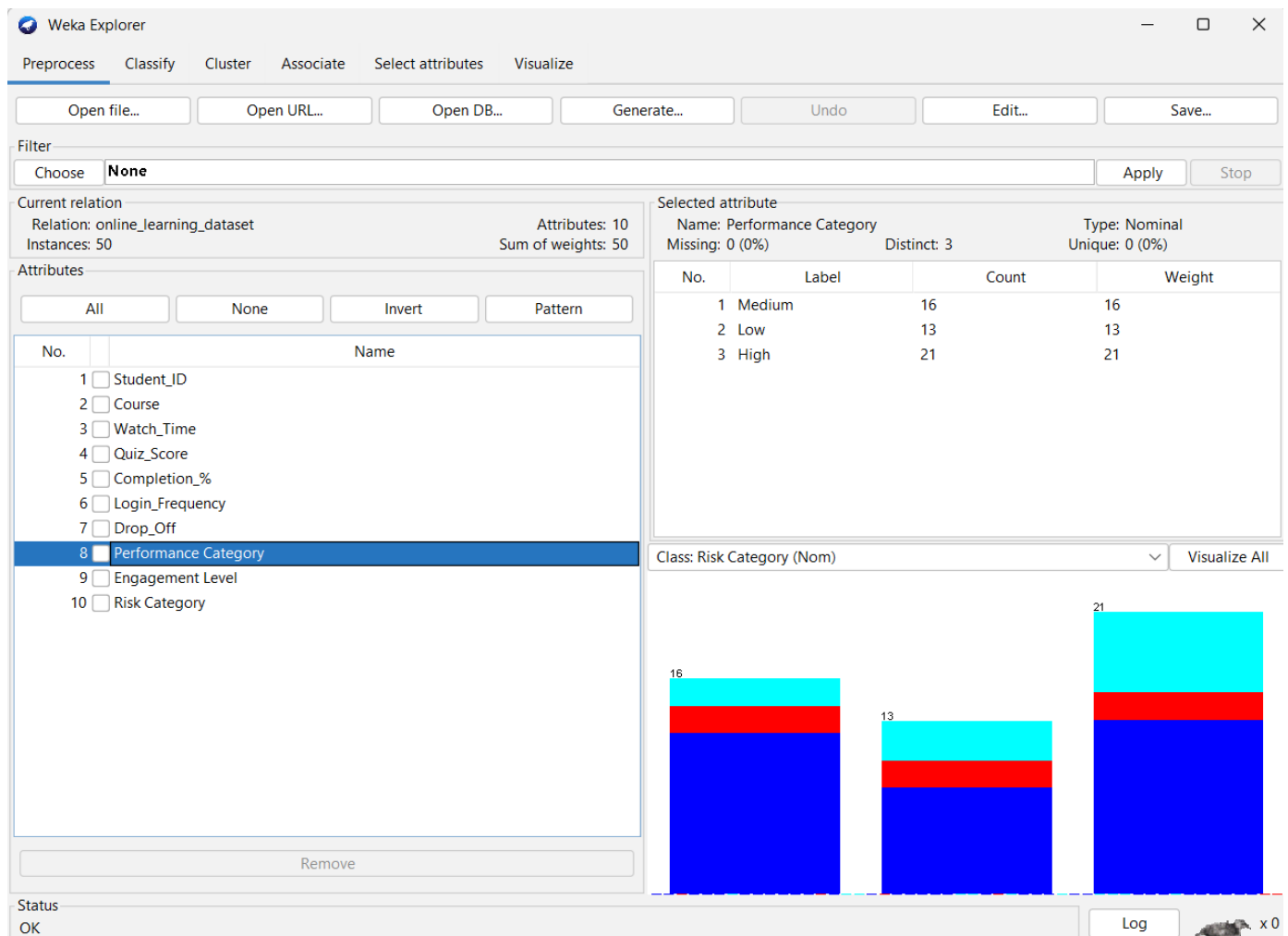
4. Methodology

The methodology of this project involves applying multiple data mining techniques using WEKA to analyze student behaviour in an online learning environment. The process includes data preprocessing, followed by classification, clustering, association rule mining, and visualization to extract meaningful insights.

4.1 Data Preprocessing

Before applying any data mining techniques, the dataset was prepared to ensure accuracy and efficiency.

- The dataset was loaded into WEKA Explorer
- The dataset was checked for missing or inconsistent values
- Numerical attributes such as Watch_Time, Quiz_Score, Completion_%, and Login_Frequency were converted into categorical ranges using the Discretize filter
- This step improved compatibility with algorithms like Apriori and enhanced model performance





4.2 Classification using Naive Bayes

Naive Bayes classifier was used to predict the Performance Category of students.

- The Classify tab in WEKA was used
- Naive Bayes algorithm was selected
- The class attribute was set as Performance Category
- A train-test split (66%) was applied to evaluate the model

Working:

- The algorithm calculates the probability of each class based on input features
- It assigns the class with the highest probability

Outcome:

- The model achieved high accuracy (~94%)
- It showed that quiz score and engagement-related attributes are strong predictors of performance

Choose NaiveBayes

Test options

☐ Use training set

☐ Supplied test set

Set...

☐ Cross-validation Folds 10

☒ Percentage split % 66

More options...

(Nom) Performance Category

Start

Stop

Result list (right-click for options)

18:32:30 - bayes.NaiveBayes

Classifier output

	3.0	4.0	7.0
Medium Risk			
[total]	19.0	16.0	24.0

Time taken to build model: 0 seconds

=== Evaluation on test split ===

Time taken to test model on test split: 0 seconds

=== Summary ===

Correctly Classified Instances	16	94.1176 %
Incorrectly Classified Instances	1	5.8824 %
Kappa statistic	0.9096	
Mean absolute error	0.0512	
Root mean squared error	0.147	
Relative absolute error	11.6576 %	
Root relative squared error	31.3215 %	
Total Number of Instances	17	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	1.000	0.077	0.800	1.000	0.889	0.859	1.000	1.000	Medium
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Low
	0.875	0.000	1.000	0.875	0.933	0.887	1.000	1.000	High
Weighted Avg.	0.941	0.018	0.953	0.941	0.942	0.914	1.000	1.000	

=== Confusion Matrix ===

a b c <-- classified as

4 0 0 | a = Medium

0 5 0 | b = Low

1 0 7 | c = High

4.3 Clustering using Hierarchical Clustering

Hierarchical clustering was applied to group students based on similarity.

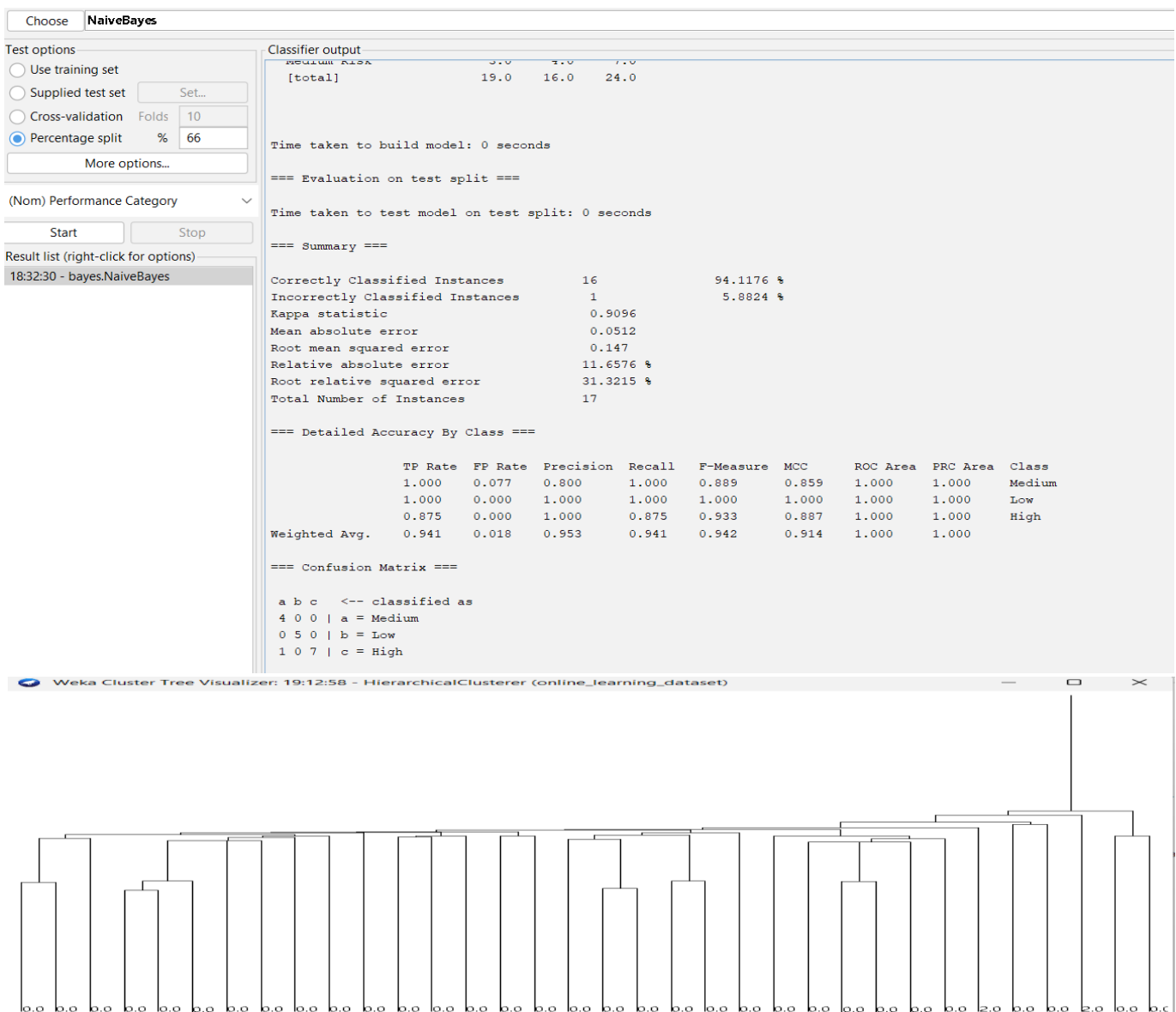
- The Cluster tab was used in WEKA
- Hierarchical Clusterer algorithm was selected
- The model grouped students into clusters without predefined labels

Working:

- It builds a hierarchy (tree-like structure) of clusters
- Similar students are grouped together based on distance measures

Outcome:

- Students were grouped into categories such as highly engaged, moderately engaged, and low engaged
- Helped in identifying different learning patterns





4.4 Association Rule Mining using Apriori

Apriori algorithm was used to find relationships between different attributes.

- Data was first discretized in the Preprocess tab
- The Associate tab was used to select Apriori algorithm
- Parameters were set:
 - Minimum Support = 0.2
 - Minimum Confidence = 0.9
 - Number of Rules = 10

Working:

- The algorithm finds frequent itemsets
- Generates rules based on support and confidence

Outcome:

- Extracted meaningful rules such as:
 - High Quiz Score → High Performance
 - Low Login Frequency → Drop Off
 - High Engagement → Low Risk

```
Associator
Choose Apriori -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1 -S -1.0 -c -1

Start Stop

Result list (right-click for ...)
19:32:40 - Apriori
19:33:47 - Apriori

Associator output
Completion_5
Login_Frequency
Drop_Off
Performance Category
Engagement Level
Risk Category

=== Associator model (full training set) ===

Apriori
=====

Minimum support: 0.2 (10 instances)
Minimum metric <confidence>: 0.9
Number of cycles performed: 16

Generated sets of large itemsets:

Size of set of large itemsets L(1): 15
Size of set of large itemsets L(2): 14
Size of set of large itemsets L(3): 3

Best rules found:
1. Engagement_Level=High 25 ==> Risk_Category=Low Risk 25 <conf:(1)> lift:(1.52) lev:(0.17) [8] conv:(8.5)
2. Drop_Off=No Engagement_Level=High 15 ==> Risk_Category=Low Risk 15 <conf:(1)> lift:(1.52) lev:(0.1) [5] conv:(5.1)
3. Watch_Time='(8.6-10.5)' 13 ==> Risk_Category=Low Risk 13 <conf:(1)> lift:(1.52) lev:(0.09) [4] conv:(4.42)
4. Completion_5='(-inf-27.9)' 11 ==> Drop_Off=Yes 11 <conf:(1)> lift:(1.92) lev:(0.11) [5] conv:(5.28)
5. Risk_Category=Medium Risk 11 ==> Engagement_Level=Medium 11 <conf:(1)> lift:(2.78) lev:(0.14) [7] conv:(7.04)
6. Watch_Time='(4.8-6.7)' 10 ==> Engagement_Level=Medium 10 <conf:(1)> lift:(2.78) lev:(0.13) [6] conv:(6.4)
7. Watch_Time='(4.8-6.7)' 10 ==> Risk_Category=Medium Risk 10 <conf:(1)> lift:(4.55) lev:(0.16) [7] conv:(7.8)
8. Watch_Time='(4.8-6.7)' Risk_Category=Medium Risk 10 ==> Engagement_Level=Medium 10 <conf:(1)> lift:(2.78) lev:(0.13) [6] conv:(6.4)
9. Watch_Time='(4.8-6.7)' Engagement_Level=Medium 10 ==> Risk_Category=Medium Risk 10 <conf:(1)> lift:(4.55) lev:(0.16) [7] conv:(7.8)
10. Watch_Time='(4.8-6.7)' 10 ==> Engagement_Level=Medium Risk_Category=Medium Risk 10 <conf:(1)> lift:(4.55) lev:(0.16) [7] conv:(7.8)
```

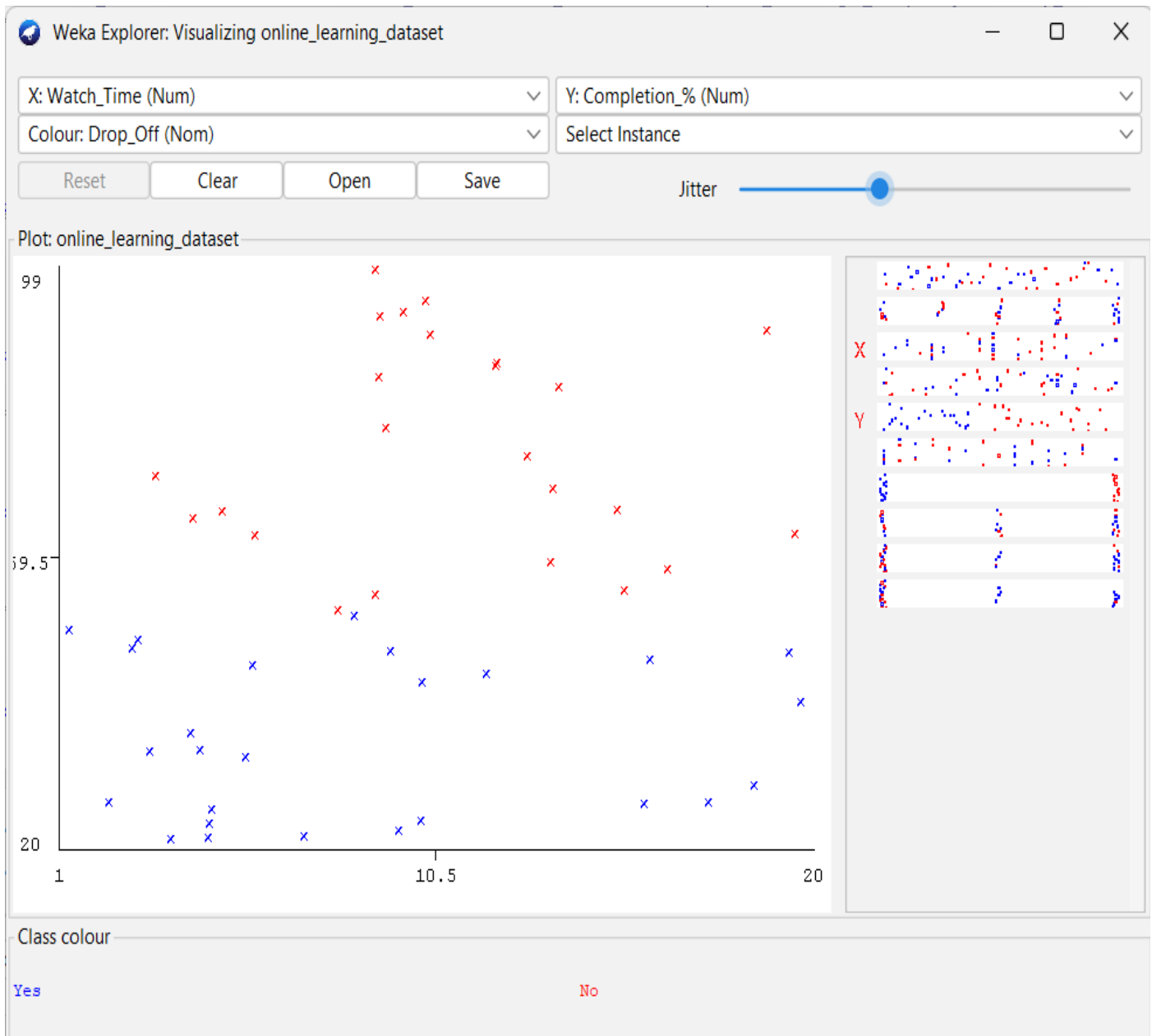
4.5 Data Visualization

Visualization was used to understand relationships between variables.

- The Visualize tab in WEKA was used
- A scatter plot was created with:
 - X-axis: Watch_Time
 - Y-axis: Completion_%

Outcome:

- A positive relationship was observed between watch time and completion percentage
- Some outliers were identified, indicating unusual student behaviour





4.6 Overall Workflow

1. Load dataset into WEKA
2. Perform preprocessing and discretization
3. Apply classification (Naive Bayes)
4. Apply clustering (Hierarchical)
5. Apply association rule mining (Apriori)
6. Visualize relationships between variables
7. Interpret results and extract insights

5. Results and Analysis

- The application of multiple data mining techniques provided a comprehensive understanding of student behaviour.
- Classification results demonstrated high predictive accuracy, confirming the reliability of the model.
- Clustering revealed natural groupings among students, helping in identifying different learning patterns.
- Association rules uncovered strong relationships between attributes, offering deeper insights into behavioural trends.
- Visualization further supported these findings by clearly showing correlations between key variables.

Overall, the combined analysis provides a holistic view of how students interact with online learning platforms.

6. Key Insights

- Students with higher quiz scores consistently perform better, indicating that assessment performance is a strong indicator of overall success.
- Increased engagement, such as regular logins and higher watch time, is directly linked to better course completion rates.
- Students with low login frequency are more likely to drop out, highlighting the importance of consistent participation.
- There exists a positive relationship between watch time and completion percentage, showing that active involvement improves outcomes.
- Different groups of students exhibit distinct learning behaviours, which can be addressed through personalized learning strategies.



7. Advantages

- The use of WEKA makes implementation simple and accessible without requiring extensive programming knowledge.
- Multiple data mining techniques can be applied within a single platform, providing a comprehensive analysis.
- The project helps in understanding complex datasets and extracting meaningful insights efficiently.
- Visualization tools enhance interpretation and make results easier to understand.

8. Limitations

- The dataset used is relatively small, which may limit the generalization of results.
- The accuracy of the analysis depends heavily on the quality and completeness of the data.
- Real-world online learning environments may involve more complex factors not captured in this dataset.
- Some assumptions made by algorithms (e.g., independence in Naive Bayes) may not fully reflect real scenarios.

9. Future Scope

- The project can be extended by using larger and real-time datasets to improve accuracy and reliability.
- Advanced machine learning techniques such as Random Forest or Neural Networks can be applied for better predictions.
- Integration with real-time dashboards can help in continuous monitoring of student performance.
- Personalized recommendation systems can be developed to guide students based on their learning patterns.

10. Conclusion

This project successfully demonstrates the application of data mining techniques to analyze online learning behaviour using WEKA. By applying classification, clustering, association rule mining, and visualization, valuable insights were extracted from the dataset. The results highlight the importance of engagement, quiz performance, and consistent participation in determining student success.

The study shows that data mining can play a significant role in improving online education systems by identifying patterns, predicting outcomes, and supporting data-driven decisions. With further enhancements and real-world implementation, such systems can greatly contribute to improving student learning experiences and reducing dropout rates.